Transformer

This week, I studied Transformer networks and applied them to a cardiovascular disease prediction task. Transformer is a deep learning architecture originally designed for natural language processing, but it has recently been adapted for tabular data classification due to its powerful self-attention mechanism, which can capture complex relationships between different features simultaneously. In this project, I used Transformer to predict whether a patient has cardiovascular disease based on 11 health indicators, including age, blood pressure, cholesterol level, glucose level, and lifestyle factors such as smoking, alcohol intake, and physical activity.

During implementation, I first preprocessed the dataset by removing unnecessary identifier columns, handling outliers in blood pressure readings, and normalizing all features using StandardScaler. To verify the preprocessing results, I visualized the distribution of each feature and analyzed the target variable balance, confirming that the data was correctly formatted and that the dataset was well-balanced with approximately 50% positive cases. Subsequently, I built a Transformer model with multi-head self-attention layers and fully connected layers, adding Dropout layers to prevent overfitting. The output layer uses sigmoid activation for binary classification. The model was trained using the Adam optimizer with binary cross-entropy as the loss function and accuracy as the evaluation metric, and early stopping was implemented to prevent overfitting.

After training, the model achieved a test accuracy of 73.22% and an AUC of 0.7991. The confusion matrix shows the model correctly identified 5,390 healthy patients and 4,679 diseased patients, with 1,555 false positives and 2,128 false negatives. The classification report demonstrates balanced performance across both classes, with F1-scores of 0.75 for healthy patients and 0.72 for diseased patients. These results demonstrate the effectiveness of Transformer in learning complex interactions between health indicators and providing reliable cardiovascular disease risk prediction.